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A Study of the Palaeomagnetism of the Bushveld Gabbro†

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Abstract

Data are presented on the natural remanent magnetization at five sites of the Main Gabbro zone of the Bushveld Igneous Complex. It is found that the directions of N.R.M. at the five sites would agree more closely if the rock at each site were rotated about the strike of the pseudo-stratification so as to make its dip zero. This indicates that the gabbro became magnetized before the crust subsided to form the basin in which the Complex now lies. The same results show that the N.R.M. of the gabbro has been stable since before the crustal subsidence. Stability of the N.R.M. is further demonstrated by the highly consistent directions of magnetization making a large angle with the present field. After correction for dip, and on the assumption that the gabbro was magnetized in the field of a geocentric dipole, the mean north magnetic pole from the five sites lies at 23° N. 36° E. The age of the Complex is close to $2{\cdot}0 \times 10^9$ years. The palaeomagnetic results show that the crust supported the load due to the gabbro intrusion without subsidence for at least the time taken by this intrusion to cool to the Curie temperature of its ferrimagnetic constituents. A lower limit to this time is estimated, on reasonable assumptions, to be of the order of 10^5 years, so that the South African crust 2×10^9 years ago appears to have had very considerable strength.

§ 1. Introduction

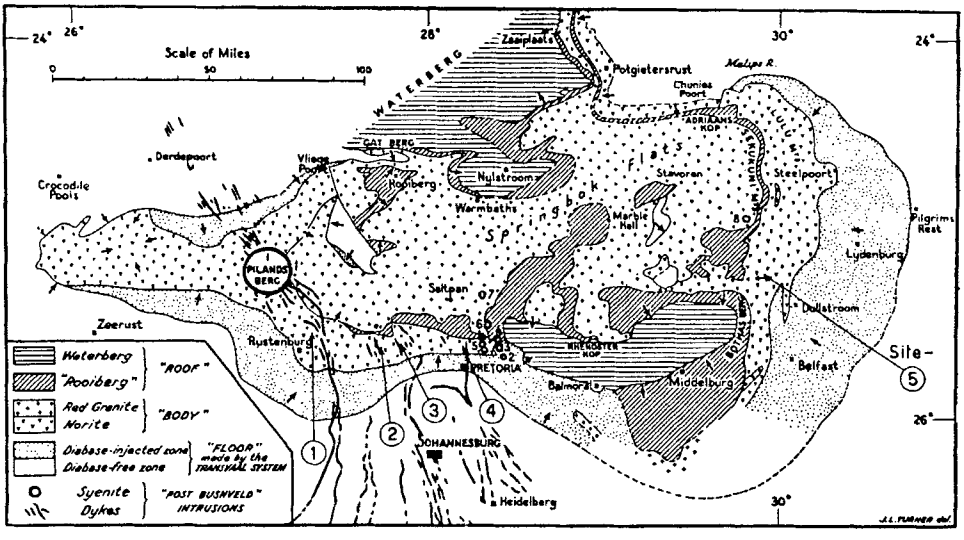
The Bushveld Igneous Complex is one of the largest systems of intrusive rocks known. Its main features are described by du Toit (1954). Figure 1 shows its general structure, as given by du Toit. A basin, some 200 miles long from east to west and about 90 miles from north to south, is filled principally with gabbro (often described as norite, as in fig. 1) and red granite, the gabbro being exposed round the flanks of the basin with the granite in the centre. The total area of gabbroic rock is nearly 20 000 sq. miles. The metamorphosed sediments of the Transvaal System which surround and underlie the gabbro, show systematic dips towards the centre of the Complex. In its lower levels the gabbro itself is differentiated into banded mineral assemblages which are inclined like the underlying sediments. Higher in the intrusion, pseudo-stratification in the undifferentiated gabbro shows similar, though less obvious, evidence of dip towards the centre of the basin. Such data have led to the conclusion that the intrusion caused an extensive subsidence of the crust at some time after crystallization was well advanced.

† Communicated by Professor P. M. S. Blackett, F.R.S.

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The age of the granite of the Complex has recently been firmly established. Schreiner (1958) used rubidium–strontium decay measured in total rock and separated mineral samples. He gives the age 1920 million years for the Bushveld granite, using half-life $5{\cdot}0 \times 10^{10}$ years, with 99% confidence limits $\pm$ 130 million years. Nicolaysen *et al.* (in the press) used rubidium–strontium and uranium–lead decay in various minerals from the Bushveld granite and give a mean age of 1950 ± 150 million years. A specimen of biotite-bearing pyroxenitic gabbro from the Merensky Reef in the differentiated zone of the gabbro was found to be 2050 ± 50 million years old (Nicolaysen *et al.*). The Bushveld Complex is therefore one of the best dated of ancient rock systems. The age of the gabbro must be close to $2{\cdot}0 \times 10^9$ years. The known age, together with the relative accuracy with which strike and dip could be defined round the basin, led to the palaeomagnetic study here reported.

Fig. 1



Simplified geological map of the Bushveld Igneous Complex, after A. L. du Toit. The five sites sampled in the gabbro are indicated.

§ 2. Sampling and Measurement

Samples were cores of diameter one-half inch drilled from the rock at the site by means of a portable field drill fitted with a diamond coring crown. Before removal each core was oriented in one of several ways. Most of the surface cores were oriented with reference to the sun's azimuth, some with reference to survey beacons and a few from a magnetic compass, which was not appreciably deflected by the gabbro. The underground cores from chrome mines were oriented relative to the mine survey networks. From one to four specimens one-half inch long were cut from each core in the laboratory, the number being two in nearly all cases.

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The natural remanent magnetization (N.R.M.) of each specimen was measured in direction and intensity by means of an electromagnetic 'spinner' type magnetometer. The specimens from the Bon Accord quarries (Site 4) were measured with the magnetometer used for the study of the Pilanesberg dykes (Gough 1956). All other specimens were measured with a new and much improved spinner magnetometer which will be described elsewhere.

Mean directions were found for each core from the specimens (normally two) cut from it, and the unit in the statistical data and in the stereograms of fig. 2 is the mean direction for a core.

§ 3. Field Work

It has been found that, in general, rock samples from the surface in South Africa are of little or no use for palaeomagnetic studies (Gough 1956, Graham and Hales 1957). Attention was therefore confined to exposures of fresh rock in excavations. A set of cores from a quarry in the Main Gabbro zone at Bon Accord, north of Pretoria (see fig. 1, Site 4) gave north-seeking poles directed near the downward vertical, as shown in fig. 2. This result immediately raised the question whether the directions of N.R.M. should be corrected for the dip of the gabbro, that is, whether the N.R.M. observed had been acquired with the rock in its present attitude or in some other. The obvious way to answer this question was to sample the formation elsewhere where its strike and dip were as widely different as possible from those at Bon Accord. Cores were therefore collected from chrome mines in the north-east and north-west flanks of the basin. No undifferentiated gabbro or norite was exposed in these mines, and the pyroxenite cores were found to be feebly magnetized with scattered directions. A set of cores from a dam site (Site 5) near Roossenekal, in the eastern flank of the Main Gabbro zone showed close grouping of directions of N.R.M. Thereafter sampling was confined to exposures of the Main Gabbro zone. Five sites were sampled, including the two already mentioned. They are marked in fig. 1. Site 1 is a group of three quarries near Rustenburg, and Sites 2 and 3 are quarries near Brits, all in the Main Gabbro. Careful search failed to discover any suitable exposures of gabbro in the west, north-west or north-east flanks of the basin.

Table 1 gives the positions of the five sites, and the strike and dip of the pseudo-stratification at each. The determinations of dip and strike presented problems because the rock at this horizon is homogeneous and shows no bands of mineral assemblages such as produce visible layering lower in the gabbro. Lenticular xenoliths of quartzite, which are oriented parallel to the pseudo-stratification, were well exposed only in one quarry. The rock develops natural fracture planes on weathering which follow the pseudo-stratification, and where such planes were freshly exposed in quarry excavations they were used for determination of dip and strike. Similar fracture surfaces exposed as outcrops at the surface were found

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to be more or less rounded by weathering, and gave systematically low dips. They were therefore used only for supplementary data on the strike. Blasting in the quarries produces splitting along parallel planes which tend to follow the pseudo-stratification. Such blasting fracture surfaces were used in the same way as natural fracture planes for measurement of strike and dip.

Table 1. Positions of the Sites and Strike and Dip at each

Site No.	Description	Latitude (°S.)	Longitude (°E.)	Strike (°E. of N.)	Dip (°)
1	Three quarries near Bospoortdam, Rustenburg district	25·54	27·35	152 ± 7	11 ± 1 E.NE.
2	Quarry at Rooikoppies, Brits district	25·62	27·74	91 ± 1	9·9 ± 0·7 N.
3	Quarry at Elandsfontein, Brits district	25·62	27·84	107 ± 4	12 ± 1 N.
4	Two quarries at Bon Accord, Pretoria district	25·62	28·21	90 ± 5	30·5 N.
5	Dam site near Roossenekal	25·27	29·95	0	13·7 ± 0·7 W.

Table 1 shows that except at Site 4, dips were all small. Individual measurements of strike and dip showed variations which are reflected in the standard errors of the mean values given in table 1.

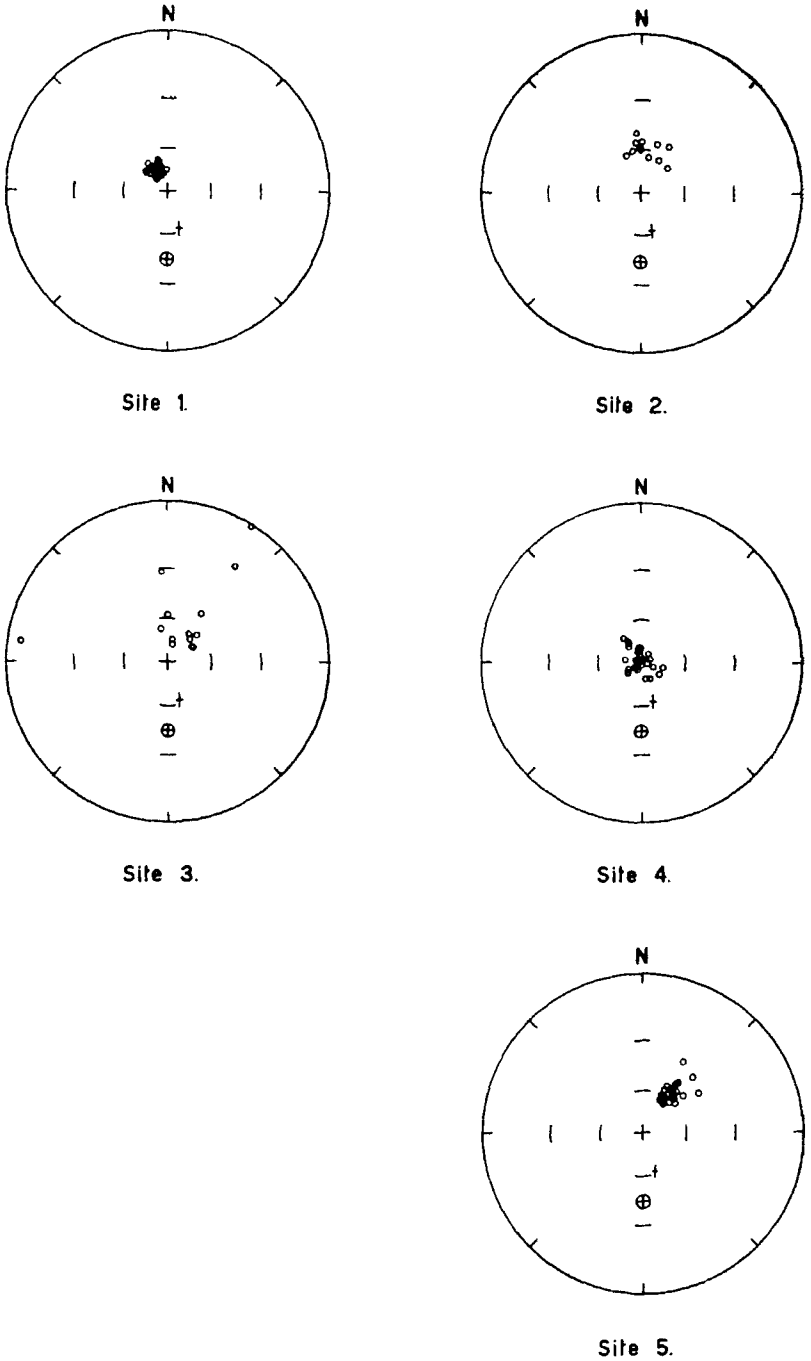
§ 4. Natural Remanent Magnetization of the Bushveld Gabbro

The directions of N.R.M. are represented stereographically in fig. 2. All the cores had north-seeking poles in the lower hemisphere of projection. It will be seen that the directions found at Sites 1, 4 and 5 were very closely grouped. Sites 2 and 3 gave less closely grouped directions. At Site 3 four cores whose N.R.M. vectors fell outside the main concentration were omitted from the statistical computation. These cores were nearer than the others to a weathered surface. At the other four sites all cores measured were included in the analysis, except two whose orientations were lost after drilling.

At Site 1 one core was drilled from gabbro into a xenolith of quartzite, and one gabbro specimen was cut from it, two quartzite, and one composite. The directions of N.R.M. were closely grouped and the mean

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Fig. 2



Directions of natural remanent magnetization at five sites in the Bushveld Gabbro. Lower hemisphere stereographic projection.

O North-seeking poles of rock cores.
⊕ Axial dipole field (south-seeking).
† Present geomagnetic field (south-seeking).

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of the four was used as one core direction in the analysis. The agreement indicates that the gabbro raised the quartzite to above the Curie temperature of its ferrimagnetic constituent. The two quartzite specimens had intensities of N.R.M. 0·13 and 0·043 of that of the gabbro specimen.

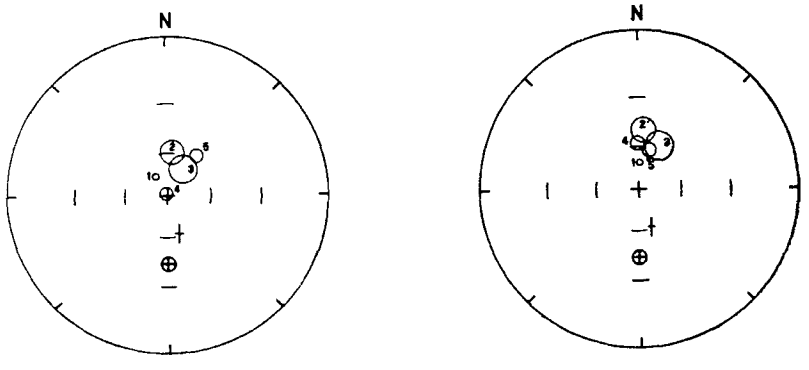
The results of the measurements of N.R.M. are given in table 2, which gives the mean direction of N.R.M. for each site, the angular radius of the cone of 99% confidence for the mean, and the parameter K, using Fisher's statistics (Fisher 1953). The confidence cones are stereographically

Table 2. Statistics of N.R.M. at Five Sites

Site	N	N'	Mean direction		α 99%	K
			Inclination	Azimuth (E. of N.)		
1	29	0	+ 73°·8	330°·0	2°·0	268·2
2	12	0	+ 59°·3	6°·9	7°·8	51·5
3	10	4	+ 68°·0	30°·2	9°·6	43·5
4	29	0	+ 87°·7	323°·9	4°·4	60·7
5	23	0	+ 55°·3	36°·8	4°·4	78·5

N = number of cores included in analysis.
N' = number of cores excluded from analysis.
α = semi-angle of cone of 99% confidence for mean direction.
$K = N - 1/N - R$ where R is the sum of N unit vectors.

Fig. 3



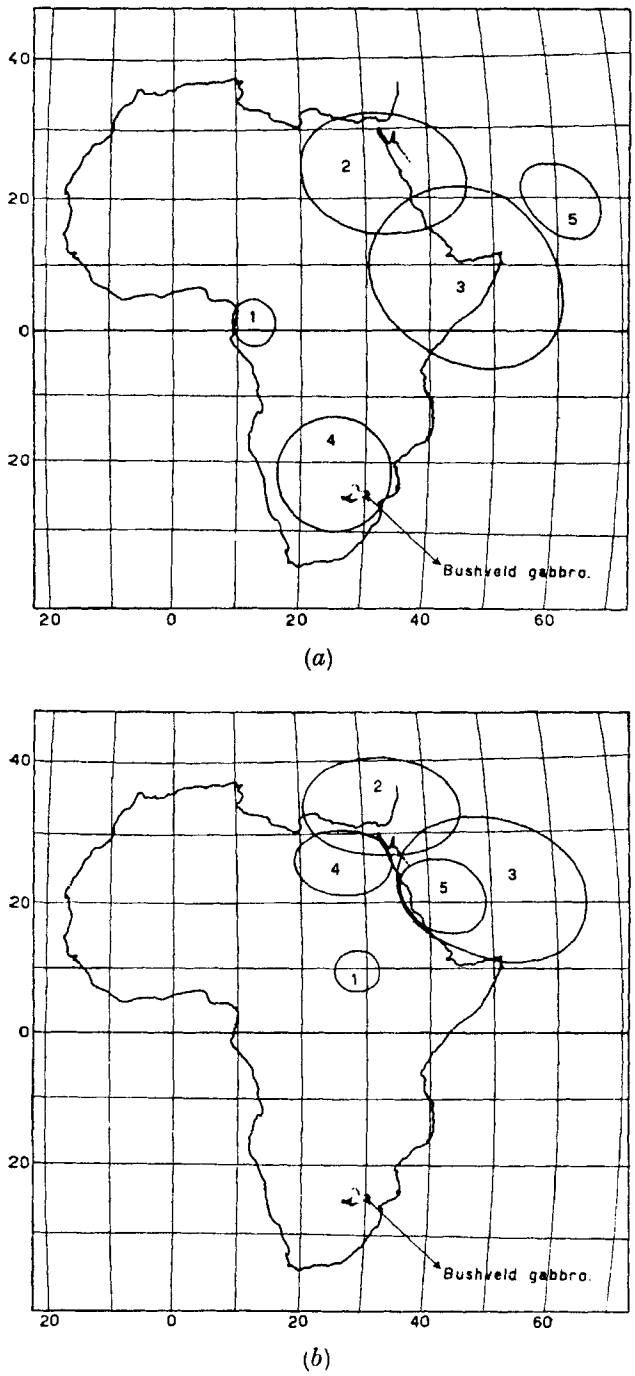
(*a*) (*b*)

Circles of 99% confidence for mean directions of magnetization at five sites in the Bushveld Gabbro. (*a*) with the rock at each site in its present attitude ; (*b*) after correction for dip of pseudo-stratification.

projected in fig. 3 (*a*). In fig. 3 (*b*) the confidence circle for each site has been moved to the position it would occupy if the gabbro at the site were rotated about the strike as axis so as to reduce the dip to zero. It will be seen that the mean directions for the five sites are more closely grouped after correction for dip than before. There are significant residual

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Fig. 4



Ovals of 99% confidence for the pole positions given by five sites in the Bushveld Gabbro, on the assumption of a geocentric dipole field. (*a*) with the rock in its present attitude at each site ; (*b*) after correction for dip.

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differences in fig. 3 (*b*), which are no doubt attributable partly to errors in the dip and strike corrections and partly to secular variation of the geomagnetic field between the dates of passage through the Curie temperature at the various sites. Sites 1 and 5 are about 160 miles apart, and it would be reasonable to suppose that the gabbro at one of them might have become magnetized hundreds of years later than that at the other.

Figure 4 (*a*) shows the oval of 99% confidence for the position of the North magnetic pole given by the data from each site, on the assumption that the observed N.R.M. was produced by the field of a geocentric dipole. Figure 4 (*b*) gives the corresponding North magnetic pole positions after correction for dip. It is clear from figs. 3 and 4 that the consistency of the data improves markedly when the pseudo-stratification is made horizontal at each site. This amounts to Graham's stability test on a large scale (Graham 1949). The simplest explanation is that the gabbro acquired its N.R.M. before the development of the crustal subsidence and that the N.R.M. has remained stable since the subsidence took place. It is possible that subsidence did not take place until after the passage of an appreciable fraction of the total time since the intrusion of the gabbro. Apart from the evidence of stability afforded by the subsidence, there is a strong argument for stability in the consistent and closely grouped N.R.M. vectors making an angle of about 120° with the present geomagnetic field. It is reasonable to suppose that the observed N.R.M. is the original thermo-remanent magnetization acquired when the gabbro cooled.

The mean positions of the North magnetic pole, derived from the five sites after correction for dip on the assumption of a geocentric dipole field, are given in table 3. The mean of the five mean pole positions with equal weights gives a North magnetic pole at latitude 23° N., longitude 36° E. There is relatively little difference between this and the pole position at the time of the intrusion of the Pilanesberg dykes.

Table 3. Mean Position for North Magnetic Pole

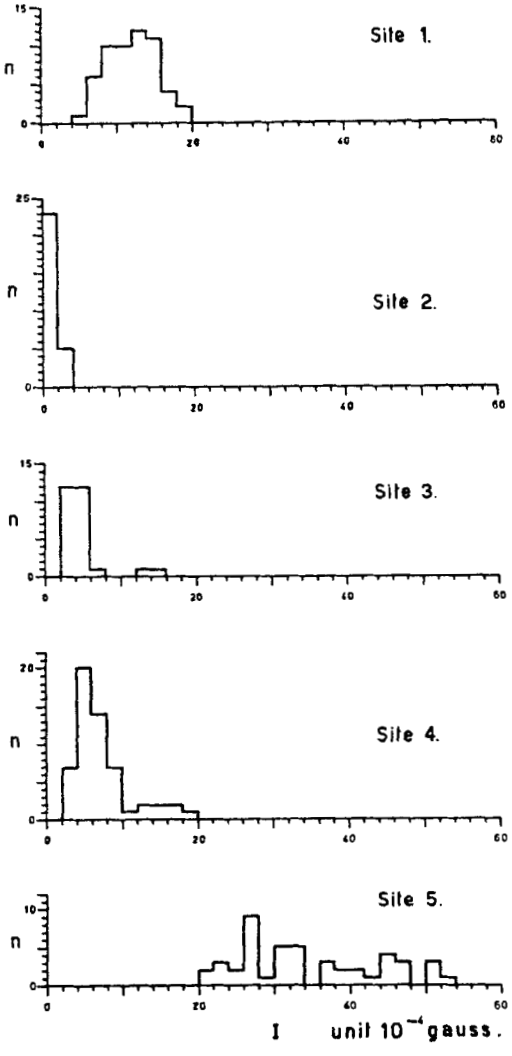
Site	Pole position	
	Latitude	Longitude
1	+10	+28
2	+34	+33
3	+23	+50
4	+26	+26
5	+20	+42
All	+23	+36

The distribution of intensity of N.R.M. at each site is shown in fig. 5. The unit here is the single specimen so that the total number of data for each site is about double that for the directions.

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The palaeomagnetic data afford evidence that the subsidence of the crust, which formed the present basin, did not develop until the rock had cooled to below the Curie temperature of its ferrimagnetic constituent.

Fig. 5



Distribution of intensities of natural remanent magnetization at five sites in the Bushveld Gabbro.

The Curie temperature for pure magnetite is 585°C and for titanomagnetites is less. It is therefore a conclusion from the work here reported that the gabbro cooled to 600°C or less before the crust subsided. It is amusing to attempt an estimate of the cooling time. The Main Gabbro forms a homogeneous body of rock which may reasonably be supposed

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to have been intruded rapidly enough so that its whole thickness was near the magma temperature at one time. In order to derive a minimum time for cooling of the Main Gabbro it will be assumed that the highly differentiated Critical Zone of the gabbro was already in place below it. Above the Main Gabbro was cap-rock whose position is now occupied by granite. The inclinations of the Transvaal System floor, of the Critical Zone, of the gabbro and of the granite are similar, and it is reasonable to suppose that the Main Gabbro formed a parallel sided slab after intrusion. If the cap-rock was much thicker than the gabbro slab, a lower limit for the cooling time would be given by taking the cap-rock above and the Critical Zone below to have been at zero temperature when the gabbro was intruded. There is in fact evidence that the Critical Zone was still at a high temperature. Du Toit (1954, p. 184) gives 15 000 feet as a representative thickness for the Main Gabbro zone. The five sites described in this paper are near the centre of the Main Gabbro. Assuming that the Main Gabbro was initially at 1000°C and had great thicknesses of rock at ordinary temperatures above and below it, an estimate can be made of the time required for it to cool to 600°C. Jaeger has calculated curves for this case, with assumed values of the thermal constants, which give

$$\text{time in years} = 0{\cdot}016D^2$$

for the temperature near the centre of such a slab to fall from 1000°C to 600°C, where D is the thickness in metres. At a point $D/20$ from the contact the corresponding time is $0{\cdot}010D^2$, so that neither convection nor the position within the slab affects the order of magnitude. With $D = 4700$ m, the time is $3{\cdot}5 \times 10^5$ years. The simplifying assumptions which do not minimize the time are (1) that the cap-rock was thick compared with the intrusion, (2) that the intrusion extended beyond the sites at least two or three times its thickness. Unless these assumptions are greatly in error it seems reasonable to conclude that the gabbro at the sampling sites took longer than a time of the order of 10^5 years to cool from 1000°C to 600°C. The crust apparently supported the gabbro for a time greater than this without subsiding into the present basin. This suggests that the crust in South Africa 2×10^9 years ago had very considerable strength. There is of course no evidence from the present work to show whether a block including the whole of the Bushveld Complex moved so as to reduce the stresses.

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We are indebted to Messrs. Oliver and Boyd and to the executors of the late Dr. A. L. du Toit for permission to reproduce as fig. 1 a geological map from Du Toit's *Geology of South Africa.*

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